Poisson Regression

Last time: Another dataset

A concerned parent asks us to investigate crime rates on college campuses. We have access to data on 81 different colleges and universities in the US, including the following variables:

- `type`: college (C) or university (U)
- `nv`: the number of crimes for that institution in the given year
- `enroll1000`: the number of enrolled students, in thousands
- `region`: region of the US C = Central, MW = Midwest, NE = Northeast, SE = Southeast, SW = Southwest, and W = West)

Last time

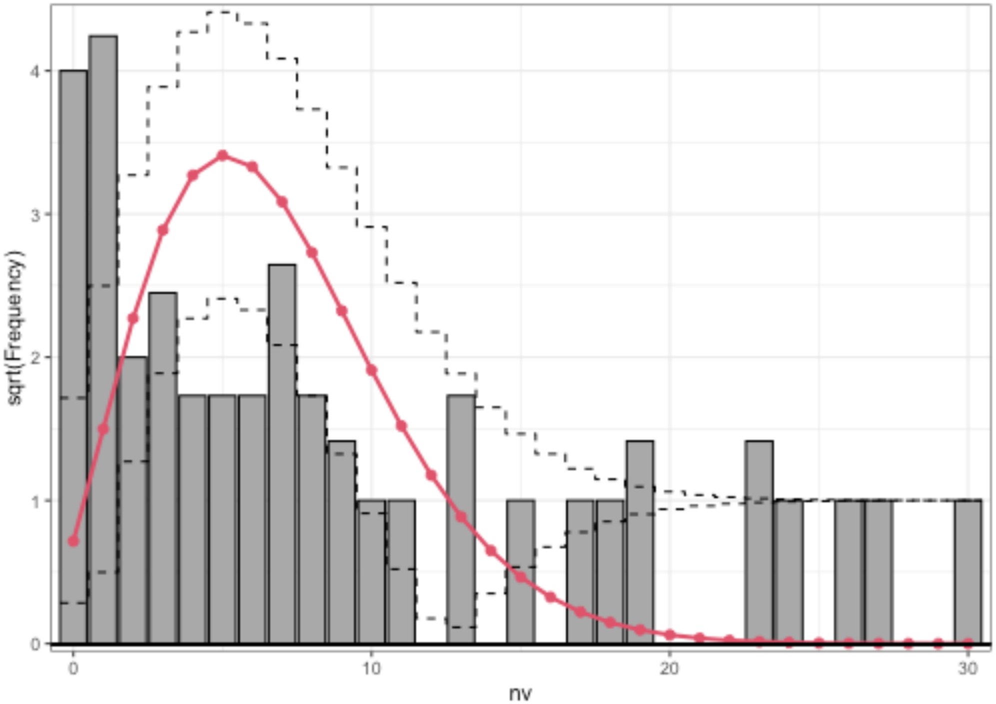
We want to know whether there are regional differences in the number of crimes on college campuses.

$$Crimes_i \sim Poisson(\lambda_i)$$

$$\log(\lambda_i) = \beta_0 + \beta_1 MW_i + \beta_2 NE_i + \beta_3 SE_i + \beta_4 SW_i + \beta_5 W_i$$

Last time: Rootogram

```r
m1 <- glm(nv ~ region, data = crimes, family = poisson)
rootogram(m1, style="standing")
```



H_0: our model is a good fit to the data

H_A: our model is not a good fit

Goodness of fit

Another way to assess whether our model is reasonable is with a *goodness of fit* test.

Goodness of fit test: If the model is a good fit for the data, then the residual deviance follows a χ^2 distribution with the same degrees of freedom as the residual deviance

```
...
## Residual deviance: 621.24  on 75  degrees of freedom
## AIC: 851.12
...
```

χ^2_{75} : mean 75

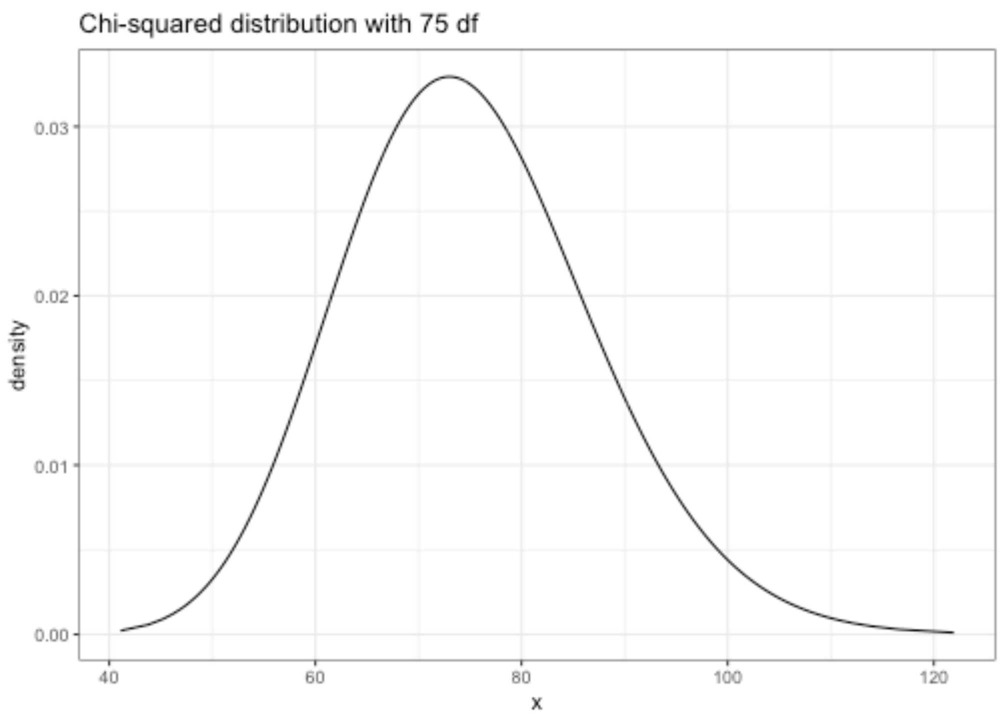
variance 150

Residual deviance = 621.24, df = 75

> How likely is a residual deviance of 621.24 if our model is correct?

Goodness of fit

Residual deviance = 621.24, df = 75




```
pchisq(621.24, df=75, lower.tail=F)
```

```
## [1] 5.844298e-87
```

≈ 0

Goodness of fit

Goodness of fit test: If the model is a good fit for the data, then the residual deviance follows a χ^2 distribution with the same degrees of freedom as the residual deviance

Residual deviance = 621.24, df = 75

```
pchisq(621.24, df=75, lower.tail=F)
```

```
## [1] 5.844298e-87
```

So our model might not be a very good fit to the data.

Potential issues with our model

later this week

- The Poisson distribution might not be a good choice
- There may be additional factors related to the number of crimes which we are not including in the model

today

Offsets

We will account for school size by including an **offset** in the model:

$$\log(\lambda_i) = \beta_0 + \beta_1 MW_i + \beta_2 NE_i + \beta_3 SE_i + \beta_4 SW_i + \beta_5 W_i + \underbrace{\log(Enrollment_i)}_{\text{offset term}}$$

(no β!)

always a log

Motivation for offsets

We can rewrite our regression model with the offset:

$$\log(\lambda_i) = \beta_0 + \beta_1 MW_i + \beta_2 NE_i + \beta_3 SE_i + \beta_4 SW_i + \beta_5 W_i + \log(Enrollment_i)$$

$$\log(\lambda_i) - \log(Enrollment_i) = \beta_0 + \beta_1 MW_i + \cdots + \beta_5 W_i$$

$$\Rightarrow \log\Big(\underbrace{\frac{\lambda_i}{Enrollment_i}}_{\text{rate of crimes}}\Big) = \beta_0 + \beta_1 MW_i + \cdots + \beta_5 W_i$$

(average # of crimes per 1000 students)

$\Rightarrow$ can now interpret βs in terms of rates, not raw counts

Motivation for offsets

We can rewrite our regression model with the offset:

$$\log(\lambda_i) = \beta_0 + \beta_1 MW_i + \beta_2 NE_i + \beta_3 SE_i + \beta_4 SW_i + \beta_5 W_i + \log(Enrollment_i)$$

Response: still $Crimes_i \sim Poisson(\lambda_i)$

(random component stays the same)

only interpretation changes (interpret βs in terms of rates)

Fitting a model with an offset

Same response!

```
m2 <- glm(nv ~ region, offset = log(enroll1000),
          data = crimes, family = poisson)
summary(m2)
```

```
...
## regionSE      0.87237    0.15313   5.697 1.22e-08 ***
## regionSW      0.50708    0.18507   2.740  0.00615 **
## regionW       0.20934    0.18605   1.125  0.26053
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
##
## (Dispersion parameter for poisson family taken to be 1)
...
```

- The offset doesn't show up in the output (because we're not estimating a coefficient for it)

Fitting a model with an offset

$$\log(\widehat{\lambda}_i) = -1.30 + 0.10MW_i + 0.76NE_i + 0.87SE_i + 0.51SW_i + 0.21W_i + \log(Enrollment_i)$$

> How would I interpret the intercept -1.30?

-1.30 = log crime rate (log average # crimes per 1000 students) for central schools

$e^{-1.30} = 0.27$ estimate 0.27 crimes per 1000 students for central schools

$e^{0.1} \approx 1.1$ The crime rate for MW schools is about 10% higher than for central schools

When to use offsets

Offsets are useful in Poisson regression when our counts come from groups of very different sizes (e.g., different numbers of students on a college campus). The offset lets us interpret model coefficients in terms of rates instead of raw counts.

> With your neighbor, brainstorm some other data scenarios where our response is a count variable, and an offset would be useful. What would our offset be?

Inference

$$\log(\lambda_i) = \beta_0 + \beta_1 MW_i + \beta_2 NE_i + \beta_3 SE_i + \beta_4 SW_i + \beta_5 W_i + \log(Enrollment_i)$$

Our concerned parent wants to know whether the crime rate on campuses is different in different regions.

> What hypotheses would we test to answer this question?

$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$

H_A: at least one of $\beta_1, \ldots, \beta_5 \neq 0$

Drop-in-deviance test

Full model:

$$\log(\lambda_i) = \beta_0 + \beta_1 MW_i + \beta_2 NE_i + \beta_3 SE_i + \beta_4 SW_i + \beta_5 W_i \\ + \log(Enrollment_i)$$

Reduced model:

$$\log(\lambda_i) = \beta_0 + \log(Enrollment_i)$$

Drop-in-deviance test

```
m2 <- glm(nv ~ region, offset = log(enroll1000),
          data = crimes, family = poisson)
summary(m2)
```

```
...
##     Null deviance: 491.00  on 80  degrees of freedom
## Residual deviance: 433.14  on 75  degrees of freedom
## AIC: 663.02
##
...
```

> What is my test statistic?

$G = 491 - 433.14 = 57.86$

Under H_0, $G \sim \chi^2_5$

Drop-in-deviance test

```
m2 <- glm(nv ~ region, offset = log(enroll1000),
          data = crimes, family = poisson)
summary(m2)
```

```
...
##     Null deviance: 491.00  on 80  degrees of freedom
## Residual deviance: 433.14  on 75  degrees of freedom
## AIC: 663.02
##
...
```

$$G = 491 - 433.14 = 57.86$$

```
pchisq(57.86, df=5, lower.tail=F)
```

```
## [1] 3.361742e-11
```

Inference

$$\log(\lambda_i) = \beta_0 + \beta_1 MW_i + \beta_2 NE_i + \beta_3 SE_i + \beta_4 SW_i + \beta_5 W_i + \log(Enrollment_i)$$

Now our concerned parent wants to know about the difference between Western and Central schools. They would like a "reasonable range" of values for the difference between the regions.

> How would we construct a "reasonable range" of values for this difference?

confidence interval (for β_5)

Confidence intervals

$$\log(\lambda_i) = \beta_0 + \beta_1 MW_i + \beta_2 NE_i + \beta_3 SE_i + \beta_4 SW_i + \beta_5 W_i + \log(Enrollment_i)$$

```
...
## (Intercept) -1.30445    0.12403 -10.517  < 2e-16 ***
## regionMW     0.09754    0.17752   0.549  0.58270
## regionNE     0.76268    0.15292   4.987 6.12e-07 ***
## regionSE     0.87237    0.15313   5.697 1.22e-08 ***
## regionSW     0.50708    0.18507   2.740  0.00615 **
## regionW      0.20934    0.18605   1.125  0.26053
## ---
...
```

95% confidence interval for β_5:

$$0.20934 \pm 1.96(0.186)$$

$$= (-0.16, 0.58)$$

Class activity

Work on the class activity (handout).

Class activity

```
               Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.304617   0.102981   2.958   0.0031 **
femWomen    -0.224594   0.054613  -4.112 3.92e-05 ***
marMarried   0.155243   0.061374   2.529   0.0114 *
kid5        -0.184883   0.040127  -4.607 4.08e-06 ***
phd          0.012823   0.026397   0.486   0.6271
ment         0.025543   0.002006  12.733  < 2e-16 ***
```

> We are interested in the relationship between prestige and the number of articles published, after accounting for other factors. What confidence interval should we make?

Class activity

$$Articles_i \sim Poisson(\lambda_i)$$

$$\log(\lambda_i) = \beta_0 + \beta_1 Female_i + \beta_2 Married_i + \beta_3 Kids_i + \beta_4 Prestige_i + \beta_5 Mentor_i$$

> Do I need an offset for this model?